Assignment #1 - Solutions

```
library(spdep)
```

1)

Consider the following data, which represent the locations of bowhead whales. Basically, the locations of a number of bowhead whales (each of which had been tagged with a radio transmitter), were noted within a region in the North Pacific. Each quadrat below represents a region 20 km by 20 km.

$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 2 & 0 & 3 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{array}$$

a) Label any non-zero quadrat "black" and any zero quadrat "white". Using the R function `joincount.multi()`, conduct a z-test for H_0 : CSR vs. H_A : not CSR, based on BB, BW, and WW joins for these data. Assume first-order ("rook's case") neighborhood structure. Use binary spatial weights (`style="B"`). Report the z-statistics and p-values for each test. You do not need to use the Monte Carlo tests here.

$$\begin{array}{cccccc} B & W & B & W & B & B \\ W & W & B & B & W & W \\ B & B & B & B & W & B \\ B & W & W & B & W & B \end{array}$$

```
#create data
whales <- c(1,0,1,0,2,1,
            0,0,1,1,0,0,
            2,2,1,2,0,3,
            1,0,0,1,0,1)

#clean
whales.bw <- 1 * (whales > 0)
whales.bw[whales.bw == 1] <- "b"
whales.bw[whales.bw == 0] <- "w"
whales.bw <- as.factor(whales.bw)

#create neighborhood structure object
whales.nb <- cell2nb(nrow = 4, ncol = 6, type = "rook")
nb2listw(whales.nb, style = "B")$weights
```

```
## [[1]]
## [1] 1 1
##
## [[2]]
## [1] 1 1 1
##
```

```
## [[3]]
## [1] 1 1 1
##
## [[4]]
## [1] 1 1 1
##
## [[5]]
## [1] 1 1 1
##
## [[6]]
## [1] 1 1
##
## [[7]]
## [1] 1 1 1
##
## [[8]]
## [1] 1 1 1 1
##
## [[9]]
## [1] 1 1 1 1
##
## [[10]]
## [1] 1 1 1 1
##
## [[11]]
## [1] 1 1 1 1
##
## [[12]]
## [1] 1 1 1
##
## [[13]]
## [1] 1 1 1
##
## [[14]]
## [1] 1 1 1 1
##
## [[15]]
## [1] 1 1 1 1
##
## [[16]]
## [1] 1 1 1 1
##
## [[17]]
## [1] 1 1 1 1
##
## [[18]]
## [1] 1 1 1
##
## [[19]]
## [1] 1 1
##
## [[20]]
## [1] 1 1 1
##
```

```
## [[21]]
## [1] 1 1 1
##
## [[22]]
## [1] 1 1 1
##
## [[23]]
## [1] 1 1 1
##
## [[24]]
## [1] 1 1
##
## attr(,"mode")
## [1] "binary"
## attr(,"B")
## [1] TRUE
```

```
whales.lwB <- nb2listw(whales.nb, style = "B")
str(whales.lwB)
```

```
## List of 3
##  $ style     : chr "B"
##  $ neighbours:List of 24
##   ..$ : int [1:2] 2 7
##   ..$ : int [1:3] 1 3 8
##   ..$ : int [1:3] 2 4 9
##   ..$ : int [1:3] 3 5 10
##   ..$ : int [1:3] 4 6 11
##   ..$ : int [1:2] 5 12
##   ..$ : int [1:3] 1 8 13
##   ..$ : int [1:4] 2 7 9 14
##   ..$ : int [1:4] 3 8 10 15
##   ..$ : int [1:4] 4 9 11 16
##   ..$ : int [1:4] 5 10 12 17
##   ..$ : int [1:3] 6 11 18
##   ..$ : int [1:3] 7 14 19
##   ..$ : int [1:4] 8 13 15 20
##   ..$ : int [1:4] 9 14 16 21
##   ..$ : int [1:4] 10 15 17 22
##   ..$ : int [1:4] 11 16 18 23
##   ..$ : int [1:3] 12 17 24
##   ..$ : int [1:2] 13 20
##   ..$ : int [1:3] 14 19 21
##   ..$ : int [1:3] 15 20 22
##   ..$ : int [1:3] 16 21 23
##   ..$ : int [1:3] 17 22 24
##   ..$ : int [1:2] 18 23
##   ..- attr(*, "class")= chr "nb"
##   ..- attr(*, "call")= language cell2nb(nrow = 4, ncol = 6, type = "rook")
##   ..- attr(*, "region.id")= chr [1:24] "1:1" "2:1" "3:1" "4:1" ...
##   ..- attr(*, "cell")= logi TRUE
##   ..- attr(*, "rook")= logi TRUE
##   ..- attr(*, "sym")= logi TRUE
##  $ weights   :List of 24
##   ..$ : num [1:2] 1 1
```

```
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..- attr(*, "mode")= chr "binary"
##   ..- attr(*, "B")= logi TRUE
##  - attr(*, "class")= chr [1:2] "listw" "nb"
##  - attr(*, "region.id")= chr [1:24] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "call")= language nb2listw(neighbours = whales.nb, style = "B")
##  - attr(*, "zero.policy")= logi FALSE
```

```
#run join tests
resultB <- joincount.multi(whales.bw, whales.lwB)
resultB
```

```
##      Joincount Expected Variance z-value
## b:b    11.0000  12.5290   3.0752 -0.8719
## w:w     6.0000   6.1957   2.5526 -0.1225
## w:b    21.0000  19.2754   8.3814  0.5957
## Jtot   21.0000  19.2754   8.3814  0.5957
```

```
#find respective pvals
2 * pnorm(-abs(resultB[1,4]))   # BB
```

```
## [1] 0.3832665
```

```
2 * pnorm(-abs(resultB[2,4]))   # WW
```

```
## [1] 0.9025355
```

```
2 * pnorm(-abs(resultB[3,4]))   # BW
```

```
## [1] 0.5513636
```

The corresponding summary table is:

Test	Joincount	Expected	Variance	z-value	P-value (2-sided)
BB	11.0000	12.5290	3.0752	−0.8719	0.3832665
WW	6.0000	6.1957	2.5526	−0.1225	0.9025355
BW	21.0000	19.2754	8.3814	0.5957	0.5513636

b) Do you reject the null hypothesis of "randomness"? Interpret the results from (a).

For the BB, WW, and BW tests, we fail to reject the null hypothesis of CSR. There is no evidence against the null.

c) If you do reject the null hypotheses, what can you say about the pattern in the data?

We do not reject the null hypothesis. There is no evidence to suggest that the data is not CSR.

2)

Repeat the exercise above using a second-order ("queen's case") neighborhood structure. Comment on the differences you observe.

```
whales.nbQ <- cell2nb(nrow = 4, ncol = 6, type = "queen")
whales.lwB <- nb2listw(whales.nbQ, style = "B")

#run join tests
resultBq <- joincount.multi(whales.bw, whales.lwB)
resultBq
```

```
##      Joincount Expected Variance z-value
## b:b    18.0000  22.4203   9.9143 -1.4038
## w:w     9.0000  11.0870   6.3482 -0.8283
## w:b    41.0000  34.4928  12.9113  1.8110
## Jtot   41.0000  34.4928  12.9113  1.8110
```

```
#find respective pvals
2 * pnorm(-abs(resultBq[1,4]))   # BB
```

```
## [1] 0.1603637
```

```
2 * pnorm(-abs(resultBq[2,4]))   # WW
```

```
## [1] 0.407499
```

```
2 * pnorm(-abs(resultBq[3,4]))   # BW
```

```
## [1] 0.07014532
```

The corresponding summary table is:

Test	Joincount	Expected	Variance	z-value	P-value (2-sided)
BB	18.0000	22.4203	9.9143	−1.4038	0.1603637
WW	9.0000	11.0870	6.3482	−0.8283	0.4074990
BW	41.0000	34.4928	12.9113	1.8110	0.07014532

b) Do you reject the null hypothesis of "randomness"? Interpret the results from (a).

For the BB, WW, and BW tests, we fail to reject the null hypothesis of CSR at the 5% level. At the 10% level, we reject the null hypothesis of CSR in the BW test. There is weak evidence for non-CSR based on the BW test.

c) What can you say about the pattern in the data?

We do not reject the null hypothesis. There is no evidence to suggest that the data is not CSR. By using the "queen's case" neighborhood structure, the joincount value, variance and expected counts, and z-values were all larger. The p-values for all three tests were smaller.

3)

By hand, list out the first-order neighbors for each cell in mosaic pattern II. Compare your answers to the `str(egII.nb)`. How many did you get? Repeat the same exercise for `type="queen"` (second-order neighbors). Submit your hand drawing of the grid and the identification of each cell and its neighbors (image).

You can check your answers with:

	Col 1	Col 2	Col 3	Col 4
Row 1	1	2	3	4
Row 2	5	6	7	8
Row 3	9	10	11	12

Plot ID	First-Order	Second-Order
1	2, 5	2, 5, 6
2	1, 3, 6	1, 3, 5, 6, 7
3	2, 4, 7	2, 4, 6, 7, 8
4	3, 8	3, 7, 8
5	1, 6, 9	1, 2, 6, 9, 10
6	2, 5, 7, 10	1, 2, 3, 5, 7, 9, 10, 11
7	3, 6, 8, 11	2, 3, 4, 6, 8, 10, 11, 12
8	4, 7, 12	3, 4, 7, 11, 12
9	5, 10	5, 6, 10
10	6, 9, 11	5, 6, 7, 9, 11
11	7, 10, 12	6, 7, 8, 10, 12
12	8, 11	7, 8, 11

Table 1: First and Second-Order Neighbors by Plot ID

```
str(cell2nb(nrow = 3, ncol = 4))
```

```
## List of 12
##  $ : int [1:2] 2 5
##  $ : int [1:3] 1 3 6
##  $ : int [1:3] 2 4 7
##  $ : int [1:2] 3 8
##  $ : int [1:3] 1 6 9
##  $ : int [1:4] 2 5 7 10
##  $ : int [1:4] 3 6 8 11
##  $ : int [1:3] 4 7 12
##  $ : int [1:2] 5 10
##  $ : int [1:3] 6 9 11
##  $ : int [1:3] 7 10 12
##  $ : int [1:2] 8 11
##  - attr(*, "class")= chr "nb"
##  - attr(*, "call")= language cell2nb(nrow = 3, ncol = 4)
##  - attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "cell")= logi TRUE
##  - attr(*, "rook")= logi TRUE
##  - attr(*, "sym")= logi TRUE
```

```r
str(cell2nb(nrow = 3, ncol = 4, type = "queen"))
```

```
## List of 12
##  $ : int [1:3] 2 5 6
##  $ : int [1:5] 1 3 5 6 7
##  $ : int [1:5] 2 4 6 7 8
##  $ : int [1:3] 3 7 8
##  $ : int [1:5] 1 2 6 9 10
##  $ : int [1:8] 1 2 3 5 7 9 10 11
##  $ : int [1:8] 2 3 4 6 8 10 11 12
##  $ : int [1:5] 3 4 7 11 12
##  $ : int [1:3] 5 6 10
##  $ : int [1:5] 5 6 7 9 11
##  $ : int [1:5] 6 7 8 10 12
##  $ : int [1:3] 7 8 11
##  - attr(*, "class")= chr "nb"
##  - attr(*, "call")= language cell2nb(nrow = 3, ncol = 4, type = "queen")
##  - attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "cell")= logi TRUE
##  - attr(*, "queen")= logi TRUE
##  - attr(*, "sym")= logi TRUE
```